



Finacle CI/CD C++ Project

Runbook

Project Details

Project Name	Finacle C++ CI/CD
Project Type	Software Development
Project Version	1.0
Name of Releaser	Kennedy N. Gatimu
File Mode of Delivery	Pipeline
Prepared By	Kennedy N. Gatimu

Overview

This runbook guides you through setting up automated deployments for Finacle scripts using Azure DevOps pipelines. Assuming the respective repositories, and CI/CD pipelines exists, it covers creating service connections, Enabling and Disabling On-premises servers, and deployment steps for various file types (FI, CBC, UNISER, and USER).

Pre-requisites

- An Azure DevOps account with a project.
- An Azure Active Directory service principal with access to deploy to on-premises environment.
- Access to create and manage resources in your Azure DevOps project.
- SSH permissions to on-premises servers DC & DR 10.1.102.135-158 & 10.1.38.79-102.
- Azure service connection user (username and password, or ssh keys).

Setup

a. Creation of Azure Service Connection:

- Go to Azure Finacle project: [Summary - Overview \(mcas.ms\)](#)
- Go to Project Settings -> Service connections.
- Click New service connection.
- Choose SSH as the service type.
- Enter on premise details:
 - Host name or IP address of the remote machine.
 - Port number on the remote machine to use for connecting, 22 is the default port for ssh
 - Enter Private Key (optional) if using ssh keys, otherwise Enter Username and password.
 - Enter Service connection name

- v. Enter Description (optional)
- vi. On Security, select Grant access permission to all pipelines.
- vii. Save
- viii. Repeat for the all the servers in this range: DC & DR
10.1.102.135-158 & 10.1.38.79-102.

b. On-Premises Server Selections

From the pool of on-premises servers, you can enable or disable which servers to be used for deployment.

- i. Go to Pipelines -> Releases
- ii. Select Edit
- iii. On the production stages, select the tasks
- iv. Agent job tasks will open displaying all the servers
- v. Enable by right clicking -> Enable Selected Task(s)
- vi. Disable by right clicking -> Disable Selected Task(s)

Only the enable tasks will deploy to the selected on-premises server.

c. Deployment Steps:

a. Pull Request Generation

For each of the repositories, (11_Scripts, 43_Scripts, 50_Scripts, 54_Scripts, 56_Scripts, 99_Scripts, and Global_Scripts) do the following:

- i. Go Azure repository: [99_Scripts - Repos \(mcas.ms\)](https://mcas.ms)
- ii. Select Clone, and copy clone URL
- iii. Open terminal and cd to the directory to clone to
- iv. Using Git, clone the repository
- v. Enter git pull command and pull from the main branch
- vi. Copy the respective Finacle files including the directory:
 - a) INFENG and FRENCH for FI, UNISER, and USER
 - b) Services for CBC
- vii. Copy below sh files:
 - a) check-modified-files.sh
 - b) deploy-cbc-scripts.sh
 - c) deploy-fi-scripts.sh

- d) deploy-uniser-scripts.sh
- e) deploy-user-scripts.sh
- f) suScript.sh
- g) azure-pipelines-CBC.yml
- h) azure-pipelines-Fl.yml
- i) azure-pipelines-UNISER.yml
- j) azure-pipelines-USER.yml
- viii. Perform below git commands:
 - a) Git add
 - b) Git commit
 - c) Git checkout -b <new-branch-name>
 - d) Git push origin <new-branch-name>
- ix. Go Azure repository: [99 Scripts - Repos \(mcas.ms\)](#)
- x. Select Repos -> Pull request
- xi. Select New pull request
- xii. Create a pull request from b <new-branch-name> to main branch
- xiii. The PR needs to be approved by at least 3 approvers
- xiv. Once approved, complete the PR.

b. CI Pipeline and Build Monitoring

The above PR will trigger a build Pipeline (CI).

To monitor the progress of the build;

- i. Go Azure Pipelines: [Pipelines - Recent \(mcas.ms\)](#)
- ii. Select the right pipeline
- iii. Go to jobs. From here you can monitor the progress of the build.
- iv. To check the actual build logs, select Checking For Modified Files task.
 - a. From this task, you will see the actual modified scripts.
- v. When the build completes, it will trigger release pipeline.

c. CD Pipeline Deployment and Monitoring

The above job will trigger Release pipeline – CD.

This stage will also require atleast 3 pre-deployment approvers.

After the approval, the deployment will start.

Monitor the progress by:

- i. Go Azure Pipelines: [Releases - Pipelines \(mcas.ms\)](#)
- ii. Select the right release
- iii. Select Production stage
- iv. Go to Agent job
- v. From here, you can monitor the progress, and logs from **Run shell commands on remote machine** section.

NOTE: Repeat the above tasks for other remaining repositories.